Semester 5													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
OE-2	Open Elective 2 (NPTEL / SWAYAM / MOOC)	3	0	0	3	3	25	25	50	--	--	100
24HS301T	Engineering Economics	3	0	0	3	3	25	25	50	--	--	100
24CH301T	Mass Transfer - 1	3	0	0	3	3	25	25	50	--	--	100
24CH301P	Mass Transfer - 1 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH302T	Chemical Reaction Engineering - 1	3	0	0	3	3	25	25	50	--	--	100
24CH302P	Chemical Reaction Engineering - 1 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH303T	Process Equipment Design	3	0	0	3	3	25	25	50	--	--	100
24CH303P	Process Equipment Design Laboratory	0	0	2	2	1	--	--	--	50	50	100
PE-1	Programme Elective 1	Students must choose one course from the given basket of elective courses for study										
24CH331T	Petroleum Refining & Petrochemicals	3	0	0	3	3	25	25	50	--	--	100
24CH332T	Renewable Energy Engineering											
24CH333T	Sustainable and Green Chemistry											
	Total	**18**	**0**	**6**	**24**	**21**						

Semester 6													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
OE-3	Open Elective 3 (Inter-department, FoET)	3	0	0	3	3	25	25	50	--	--	100
24CH304T	Instrumentation and Process Control	3	0	0	3	3	25	25	50	--	--	100
24CH304P	Instrumentation and Process Control Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH305T	Chemical Reaction Engineering - 2	3	0	0	3	3	25	25	50	--	--	100
24CH305P	Chemical Reaction Engineering - 2 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH306T	Mass Transfer - 2	3	0	0	3	3	25	25	50	--	--	100
24CH306P	Mass Transfer - 2 Laboratory	0	0	2	2	1	--	--	--	50	50	100
PE-2	Programme Elective 2	Students must choose one course from the given basket of elective courses for study										
24CH334T	Piping Design	3	0	0	3	3	25	25	50	--	--	100
24CH335T	Corrosion Engineering											
24CH336T	Material Science & Engineering											
PE-3	Programme Elective 3	Students must choose one course from the given basket of elective courses for study										
24CH337T	Nano Technology & Energy Storage	3	0	0	3	3	25	25	50	--	--	100
24CH338T	Membrane Processes											
24CH339T	Environmental Engineering and Pollution Control											
	Total	**18**	**0**	**6**	**24**	**21**						

24HS301T					Engineering Economics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To introduce the fundamental principles of economics relevant to engineering decision making.
- To develop the ability to evaluate the time value of money and compare engineering alternatives.
- To understand cost concepts, depreciation, taxation and their role in project appraisal.
- To appreciate the macroeconomic and financial environment in which engineering enterprises operate.

Unit I: FUNDAMENTALS OF ENGINEERING ECONOMICS **Hours: 9**

Nature and scope of economics; engineering economics and its role in decision making; demand and supply analysis; elasticity of demand; law of returns; theory of production; economies and diseconomies of scale; market structures - perfect competition, monopoly, monopolistic competition and oligopoly; pricing policies and practices.

Unit II: COST CONCEPTS AND TIME VALUE OF MONEY **Hours: 10**

Cost concepts: fixed, variable, marginal, opportunity, sunk and incremental costs; short-run and long-run cost curves; break-even analysis; interest and interest rates; simple and compound interest; cash flow diagrams; time value of money; present worth, future worth, annual worth and capitalized cost; nominal and effective interest rates.

Unit III: ECONOMIC EVALUATION OF ALTERNATIVES **Hours: 10**

Comparison of engineering alternatives; present worth, annual worth and rate of return methods; internal rate of return and incremental analysis; benefit-cost ratio analysis; payback period; depreciation - straight line, declining balance, sum-of-years-digits and sinking fund methods; taxation and after-tax cash flow analysis; replacement analysis.

Unit IV: FINANCIAL STATEMENTS AND MACROECONOMIC ENVIRONMENT **Hours: 10**

Elements of book-keeping and accounting; balance sheet, profit and loss account and cash flow statement; ratio analysis; working capital management; introduction to national income, inflation, monetary and fiscal policy; business cycles; introduction to Indian industrial and economic policy relevant to the process industries.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall the fundamental concepts, terminology and principles of engineering economics.
CO2 - Understand the relevance of demand, supply, cost and market structure to engineering enterprises.
CO3 - Apply time value of money concepts to engineering cash flows.
CO4 - Analyse engineering alternatives using present worth, annual worth and rate of return methods.
CO5 - Evaluate depreciation, taxation and replacement decisions for engineering assets.
CO6 - Interpret financial statements and appraise the economic environment of an engineering organization.

TEXT / REFERENCE BOOKS

1. Blank, L., & Tarquin, A. (2017). *Engineering economy* (8th ed.). McGraw-Hill Education.
2. Panneerselvam, R. (2013). *Engineering economics* (2nd ed.). PHI Learning.
3. Sullivan, W. G., Wicks, E. M., & Koelling, C. P. (2018). *Engineering economy* (17th ed.). Pearson Education.
4. Thuesen, G. J., & Fabrycky, W. J. (2001). *Engineering economy* (9th ed.). Prentice Hall of India.
5. Chandra, P. (2019). *Financial management: Theory and practice* (10th ed.). McGraw Hill Education.

24CH301T					Mass Transfer - 1					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the fundamentals of mass transfer such as diffusion, mass transfer theories and mass transfer coefficients.
- To examine the various mass transfer theories.
- To apprehend the concept of transfer units.
- To apply the interphase mass transfer concept.

Unit I: INTRODUCTION TO MASS TRANSFER OPERATIONS **Hours: 12**

Fick's law of diffusion; equimolecular counter diffusion in fluids; diffusion in a stationary gas; diffusion through variable cross-sectional area; diffusion coefficient; multicomponent diffusion; diffusivity in solids; diffusion of sensitive materials: safety considerations in design and operation.

Unit II: MASS TRANSFER COEFFICIENTS **Hours: 9**

Introduction to mass transfer coefficients; correlations for convective mass transfer coefficients; mass transfer equilibrium; theories of mass transfer; local and overall mass transfer coefficients; steady state co-current and counter-current processes; material balance; stage efficiency.

Unit III: ABSORPTION **Hours: 9**

Equilibrium solubility of gases; material balance for transfer of one component; counter-current multistage operations for binary and multicomponent systems; continuous contactors; absorption with chemical reaction; concept of HTU and NTU; equipment for gas-liquid operations; calculation of the required flow rate and mass transfer area for safe scrubbing of released toxic gas.

Unit IV: ADSORPTION **Hours: 9**

Introduction; adsorption isotherms; heat of adsorption; selection of adsorbent; batch adsorption; rate of adsorption in fixed beds; adsorption equipment; adsorption dynamics; regeneration of adsorbent; thermal and pressure swing adsorption; safety issues associated with drying of volatile hydrocarbons in adsorbent beds.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the basic principles of diffusion, mass transfer theories and interphase mass transfer.
CO2 - Explain the principles of diffusion and mass transfer operations such as adsorption, absorption and stripping.
CO3 - Apply principles related to diffusion, adsorption and absorption.
CO4 - Analyze the problems related to diffusion, adsorption and absorption.
CO5 - Justify the applications of the theory learned in industrial practice regarding diffusion, adsorption and absorption.
CO6 - Design columns related to diverse mass transfer operations such as adsorption and absorption.

TEXT / REFERENCE BOOKS

1. Treybal, R. E. (2012). *Mass transfer operations* (3rd ed.). McGraw Hill Education.
2. Dutta, B. K. (2006). *Principles of mass transfer and separation processes*. Prentice Hall India Learning.
3. Narayanan, K. V. (2005). *Mass transfer: Theory and applications*. CBS Publishers.
4. Wankat, P. C. (1988). *Separations in chemical engineering: Equilibrium staged separations*. Prentice Hall.
5. Coulson, J. M., & Richardson, J. F. (2006). *Particle technology and separation processes* (5th ed.). Elsevier.
6. McCabe, W. L., Smith, J. C., & Harriott, P. (2014). *Unit operations of chemical engineering* (7th ed.). McGraw Hill Education.

24CH301P					Mass Transfer - 1 Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To understand and learn to quantify the diffusivity and mass transfer coefficient.
- To understand the principle of pressure and temperature swing adsorption.
- To understand and learn the principles of adsorption and absorption.
- To apply mass transfer theories to characterize packed beds and sieve trays for mass transfer.

LIST OF EXPERIMENTS

1. To study the absorption of carbon dioxide by aqueous sodium hydroxide solution in a packed bed column.
2. To study solid-in-air diffusion using a packed bed of spherical particles of naphthalene.
3. To determine the diffusion coefficient of a liquid in air and its variation with temperature.
4. To study the absorption of carbon dioxide by aqueous sodium hydroxide solution in a sieve tray column.
5. To study the absorption of carbon dioxide by aqueous sodium hydroxide solution in a wetted wall tower.
6. To study the batch adsorption of acetic acid by activated carbon and test the validity of adsorption isotherms.
7. To study the adsorption in a packed bed for a solid-liquid system.
8. To determine the mass transfer coefficient in interphase mass transfer.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define diffusivity, mass transfer coefficient and isotherms.
CO2 - Explain the extent of mass transfer.
CO3 - Apply principles related to diffusion, absorption and adsorption.
CO4 - Distinguish the effectiveness of packed beds, empty columns and sieve trays.
CO5 - Determine the diffusivity and mass transfer coefficient in different types of systems.
CO6 - Estimate the extent of mass transfer and the constants of different isotherms.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Mass transfer laboratory manual*. Pandit Deendayal Energy University.
2. Treybal, R. E. (2012). *Mass transfer operations* (3rd ed.). McGraw Hill Education.
3. McCabe, W. L., Smith, J. C., & Harriott, P. (2005). *Unit operations of chemical engineering* (7th ed.). McGraw-Hill.
4. Dutta, B. K. (2006). *Principles of mass transfer and separation processes*. Prentice Hall India Learning.
5. Narayanan, K. V. (2005). *Mass transfer: Theory and applications*. CBS Publishers.
6. Wankat, P. C. (1988). *Separations in chemical engineering: Equilibrium staged separations*. Prentice Hall.

24CH302T					Chemical Reaction Engineering - 1					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To provide an understanding of the basic principles and terminology of reaction kinetics.
- To acquaint students with the basic design of ideal reactors.
- To allow students to analyse rate data and thereby find the kinetic parameters of interest.
- To illustrate the concept of reactor sequencing and evaluate the performance of ideal reactors.

Unit I: BASICS OF KINETICS **Hours: 9**

Mole balances; kinetics of homogeneous reactions - rate of reaction, types of reactions, reaction mechanism, temperature and concentration dependent terms of a rate equation; searching for a reaction mechanism; rate law and stoichiometry; concentration and conversion change; introduction to batch and continuous reactor systems; sizing and characteristics of reactors.

Unit II: ANALYSIS OF RATE DATA **Hours: 12**

Collection and analysis of rate data; integral and differential methods of analysis of data; batch reactor data; variable and constant volume systems; method of initial rates; method of half-life; differential reactors; least square analysis.

Unit III: ISOTHERMAL REACTOR DESIGN **Hours: 9**

Ideal reactors: batch reactors, semi-batch reactors, continuous-flow reactors (MFR and PFR), recycle reactors; conversion and reactor sizing; design equations; applications of the design equations for continuous-flow reactors; steady and unsteady operation of ideal reactors; reactor safety.

Unit IV: REACTOR SEQUENCING **Hours: 9**

Reactors in series; reactors for parallel reactions; maximizing the desired product in parallel reactions; maximizing the desired product in series reactions; temperature and pressure effects in single and multiple reactions; adiabatic reactor design; criteria for reactor selection and its operating conditions; adiabatic reactions and thermal runaway.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Relate the basics of kinetics and basic theories to obtain the underlying mechanisms.
CO2 - Interpret and evaluate the rate data and obtain the kinetic parameters.
CO3 - Select proper design equations and perform reactor sizing for ideal reactors.
CO4 - Examine the suitable combinations of ideal reactors for optimal performance.
CO5 - Decide the reactor sequencing for single and multiple reactions towards desired products.
CO6 - Evaluate reactor systems and recommend the most suitable reactor and its operating conditions.

TEXT / REFERENCE BOOKS

1. Fogler, H. S. (2016). *Elements of chemical reaction engineering* (5th ed.). Prentice Hall.
2. Levenspiel, O. (1999). *Chemical reaction engineering* (3rd ed.). John Wiley & Sons.
3. Smith, J. M. (1988). *Chemical engineering kinetics* (3rd ed.). McGraw-Hill.
4. Schmidt, L. D. (1998). *The engineering of chemical reactions*. Oxford University Press.

24CH302P					Chemical Reaction Engineering - 1 Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To perform experiments with different ideal reactors for homogeneous liquid phase reactions to determine kinetics.
- To find the kinetic parameters using the Arrhenius theory by experimentation with different ideal reactors.
- To experiment with combinations of ideal reactors and analyse the performance optimization.
- To find the application of combinations of ideal reactors and the change in performance with the ideal reactors.

LIST OF EXPERIMENTS

1. To study a non-catalytic homogeneous liquid phase reaction in an ambient CSTR.
2. To study a non-catalytic homogeneous liquid phase reaction in an ambient straight tube PFR.
3. To study a non-catalytic homogeneous liquid phase reaction in an ambient semi-batch reactor.
4. To study a non-catalytic homogeneous liquid phase reaction in three cascaded ambient CSTRs.
5. Evaluating kinetic parameters (k0 and E) for a saponification reaction in a batch reactor.
6. To study a non-catalytic homogeneous liquid phase reaction in an ambient combined flow reactor (CSTR + PFR).
7. To perform a saponification reaction for a pseudo first order reaction in a CSTR (M ≈ 25-50).
8. To perform a saponification reaction for a variable molar ratio of reactants in a CSTR (M > 1).

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Relate the basics of kinetics and their application in practice.
CO2 - Interpret and evaluate the rate data and obtain the kinetic parameters for different reactors.
CO3 - Apply proper rate equations and compare the performance of different ideal reactors.
CO4 - Examine the combinations of ideal reactors and analyse the theoretical findings.
CO5 - Decide the reactor sequencing to optimize the volume or conversion for given combinations of reactors.
CO6 - Develop an ability to work in a group, learn report making and improve soft skills in presenting the findings.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Chemical reaction engineering laboratory manual*. Pandit Deendayal Energy University.
2. Fogler, H. S. (2016). *Elements of chemical reaction engineering* (5th ed.). Prentice Hall.
3. Levenspiel, O. (1999). *Chemical reaction engineering* (3rd ed.). John Wiley & Sons.
4. Smith, J. M. (1988). *Chemical engineering kinetics* (3rd ed.). McGraw-Hill.
5. Schmidt, L. D. (1998). *The engineering of chemical reactions*. Oxford University Press.

24CH303T					Process Equipment Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To provide a fundamental understanding of the process design of fluid and heat transport equipment.
- To develop an understanding of international standards, fabrication and testing methods.
- To execute the sizing calculations of pipes and of heat and mass transfer equipment.
- To develop the ability to design chemical reactors and storage vessels with a holistic design approach.

Unit I: DESIGN OF FLUID TRANSPORT EQUIPMENT **Hours: 12**

Process design of piping; designs based on internal and external design pressures; process design of centrifugal pumps, fans, blowers and compressors; piping networks; fundamentals of piping and instrumentation (P&I); safety issues of high-pressure and high-vacuum vessels.

Unit II: DESIGN OF HEAT TRANSPORT EQUIPMENT **Hours: 9**

Process design of heat exchangers; shell and tube heat exchangers; the design method of shell and tube heat exchangers; general mechanical design considerations; reboilers; heat-exchanger networks; safety in storage systems for thermal fluids and hot oils.

Unit III: DESIGN OF MASS TRANSPORT EQUIPMENT **Hours: 11**

Process design of tray-based distillation and absorption columns; process design and selection criteria for various internals; considerations in packed columns; process design of a column network; general mechanical design considerations; safety aspects for tall vertical towers (e.g. wind loading, structural stability, fall protection).

Unit IV: DESIGN OF CHEMICAL REACTORS **Hours: 7**

Process design of chemical storage vessels and adiabatic and isothermal reactors; process design of reactors in batch, semi-batch and continuous operations; general mechanical design considerations; safety considerations in handling reactive and pressurized chemicals.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Explain the basics of chemical process equipment design.
CO2 - Understand the international standards, fabrication and testing methods.
CO3 - Execute the sizing calculations of pipes and of heat transfer and mass transfer equipment.
CO4 - Understand the parametric interdependence and the need for a holistic design approach.
CO5 - Critically analyse and combine the elements of various designs of process equipment.
CO6 - Design the different types of equipment used in chemical process industries.

TEXT / REFERENCE BOOKS

1. Ray, S., & Das, G. (2020). *Process equipment and plant design: Principles and practices*. Elsevier Science.
2. Thakore, S. B., & Bhatt, B. I. (2018). *Introduction to process engineering and design* (2nd ed.). Tata McGraw Hill.
3. Sinnott, R. K. (2015). *Chemical engineering design: Principles, practice and economics of plant and process design* (2nd ed.). Butterworth Heinemann.
4. Coker, A. K. (2011). *Ludwig's applied process design for chemical and petrochemical plants* (Vols. 1-3, 4th ed.). Gulf Publishing Company.
5. Thakore, S. B. (2021). *Illustrated process equipment design* (2nd ed.). McGraw Hill.

24CH303P					Process Equipment Design Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To apply the basics of chemical process equipment design in a simulation.
- To understand the sizing fundamentals of the process design of equipment.
- To analyze equipment designs against different evaluation criteria for various applications.
- To provide students with hands-on training on computational tools for equipment design.

LIST OF EXPERIMENTS

1. Introduction to various process simulation software and understanding the modelling of unit operations.
2. Process calculation and simulation for mixing streams using a pipeline network.
3. Process calculation and simulation of a pumping system for liquid streams.
4. Process calculation and simulation of a compressor/blower system.
5. Process calculation and simulation of heat exchangers.
6. Process calculation and simulation of multi-effect evaporators.
7. Process calculation and simulation of distillation operation using the McCabe-Thiele method.
8. Process calculation and simulation of short-cut approaches for water-methanol separation using distillation.
9. Process calculation and simulation of rigorous approaches for water-methanol separation using distillation.
10. Process calculation and simulation of the absorption of acetic acid using water as solvent.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall the basics of computer-aided software used in process calculations.
CO2 - Explain the methodology used in design calculations.
CO3 - Apply the knowledge gained for the optimisation of process variables.
CO4 - Analyse critically for the optimum design.
CO5 - Recommend the various internals needed in a chemical process.
CO6 - Design and develop process flow diagrams.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Process equipment design laboratory manual*. Pandit Deendayal Energy University.
2. Ray, S., & Das, G. (2020). *Process equipment and plant design: Principles and practices*. Elsevier Science.
3. Thakore, S. B., & Bhatt, B. I. (2018). *Introduction to process engineering and design* (2nd ed.). Tata McGraw Hill.
4. Sinnott, R. K. (2015). *Chemical engineering design: Principles, practice and economics of plant and process design* (2nd ed.). Butterworth Heinemann.
5. Thakore, S. B. (2021). *Illustrated process equipment design* (2nd ed.). McGraw Hill.

24CH331T					Petroleum Refining and Petrochemicals					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 1)

COURSE OBJECTIVES

- To develop a fundamental understanding of units involved in petroleum refineries and the petrochemical industry.
- To understand the operation and importance of various conversion processes employed in a refinery.
- To discuss the various characterization techniques and testing methods of different petroleum fractions.
- To understand the manufacturing processes and applications of widely used petrochemicals.

Unit I: EVALUATION AND CHARACTERISATION OF CRUDE OIL AND PETROLEUM PRODUCTS **Hours: 7**

Origin and occurrence; status of Indian petroleum industries; pre-treatment of crude oil for refining; petroleum processing data - evaluation of petroleum; thermal properties of petroleum fractions; characterization of crude oil and product specification.

Unit II: PRETREATMENT AND FRACTIONATION OF CRUDE OIL **Hours: 9**

Dehydration and desalting of crudes; fractionation of petroleum through atmospheric and vacuum distillation; heating of crude - pipe still heaters; treatment techniques for impurities in various fractions.

Unit III: THERMAL AND CATALYTIC PROCESSES FOR PRODUCT UPGRADATION **Hours: 14**

Thermal and catalytic processes: thermal cracking, catalytic cracking, fluidized catalytic cracking, catalytic reforming, thermal reforming, coking technology; hydrogen processes - hydrocracking, hydrotreatment, hydrodesulphurization; alkylation and isomerisation processes.

Unit IV: PROCESS TECHNOLOGY OF PETROCHEMICAL PRODUCTS **Hours: 9**

Indian petrochemical industry overview; petrochemical industry feedstocks; synthesis gas and chemicals; petrochemicals derived from C1, C2, C3, C4 and BTX aromatics; synthetic fibres; shale gas; gas hydrates; coal bed methane; manufacturing technologies for C1, C2, C3, C4 and aromatic compounds.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Outline the crude composition, properties and characterization methods of different petroleum fractions.
CO2 - Understand the fractionation process of crude oil.
CO3 - Determine the different conversion processes of the heavier fractions by cracking and coking technology.
CO4 - Illustrate the process technology of product upgradation units and hydrogen processes.
CO5 - Review the treatment processes for preparing finished products using chemical or physical separation.
CO6 - Facilitate the process technology of various important petrochemical products.

TEXT / REFERENCE BOOKS

1. Gary, J. H., Handwerk, G. E., & Kaiser, M. J. (2007). *Petroleum refining: Technology and economics* (5th ed.). CRC Press.
2. Bhaskara Rao, B. K. (2010). *Modern petroleum refining processes* (5th ed.). Oxford & IBH.
3. Speight, J. G. (2014). *The chemistry and technology of petroleum* (5th ed.). CRC Press.
4. Nelson, W. L. (1985). *Petroleum refinery engineering* (4th ed.). McGraw-Hill.

24CH332T					Renewable Energy Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 1)

COURSE OBJECTIVES

- To know the basic scientific processes behind various renewable energy conversion technologies.
- To understand the design, working and performance analysis of renewable energy systems.
- To build engineering perspectives on efficient energy generation and grid integration.
- To appreciate the policy, economic and environmental frameworks governing renewable energy.

Unit I: SOLAR ENERGY TECHNOLOGIES **Hours: 12**

Solar radiation; photovoltaic (PV) technology; PV cell materials and fabrication techniques; solar thermal energy conversion systems; concentrating solar power (CSP) technologies; design and performance analysis of PV and CSP systems; integration of solar energy systems into the grid.

Unit II: WIND AND HYDROPOWER SYSTEMS ENGINEERING **Hours: 9**

Wind turbine technology and aerodynamics; wind resource assessment and wind farm design; hydropower system components and design; tidal and wave energy conversion mechanisms; environmental impact and sustainability considerations.

Unit III: BIOENERGY AND GEOTHERMAL ENERGY SYSTEMS **Hours: 9**

Biomass energy sources and conversion technologies; biofuels production and waste-to-energy systems; geothermal heat extraction and power generation; ground-source heat pumps and direct use applications; environmental impact and resource management.

Unit IV: SMART GRIDS, ENERGY STORAGE AND POLICY **Hours: 9**

Smart grid technologies and renewable integration; energy storage solutions - batteries, fuel cells and mechanical systems; energy policy, economics and market mechanisms; regulatory framework and incentives for renewable energy; case studies in renewable energy implementation and best practices.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Acquire in-depth expertise in the latest renewable energy technologies.
CO2 - Develop skills to design and optimize advanced renewable energy systems for grid integration.
CO3 - Critically assess and apply the latest energy storage technologies to enhance system reliability.
CO4 - Proficiently evaluate the environmental impact and sustainability of renewable energy systems.
CO5 - Understand and strategize the legal and economic frameworks for renewable energy systems.
CO6 - Conduct innovative study and advance knowledge in renewable energy engineering.

TEXT / REFERENCE BOOKS

1. Duffie, J. A., & Beckman, W. A. (2013). *Solar engineering of thermal processes* (4th ed.). John Wiley & Sons.
2. Fuller, T. F., & Harb, J. N. (2018). *Electrochemical engineering*. Wiley.
3. Lvov, S. N. (2015). *Electrochemical science and engineering*. CRC Press.
4. Boyle, G. (2012). *Renewable energy: Power for a sustainable future* (3rd ed.). Oxford University Press.
5. Kothari, D. P., Singal, K. C., & Ranjan, R. (2011). *Renewable energy sources and emerging technologies* (2nd ed.). PHI Learning.

24CH333T					Sustainable and Green Chemistry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

(Programme Elective 1)

COURSE OBJECTIVES

- To learn the fundamental philosophy and the basic importance of sustainability and green chemistry.
- To understand the principles, synthesis methods, applications and the latest developments in sustainable chemistry.
- To acquaint students with various green reaction pathways of conventional reactions with real-world applications.
- To understand the necessity for solvent replacement options and the new process development options available.

Unit I: SUSTAINABILITY AND NEED OF GREEN CHEMISTRY **Hours: 7**

Chemistry from past to future; importance of sustainability; need for green chemistry; fundamentals of sustainable development; growth, consumption and natural wealth; sustainable development at different scales.

Unit II: GREEN CHEMISTRY **Hours: 13**

Green chemistry: principles and applications; synthesis and green chemistry - micro-reactor technology, solvent-less reactions, use of green solvents; brief introduction to combinatorial chemistry and the role of AI in chemistry.

Unit III: GREEN SOLVENTS AND SOLVENT SELECTION **Hours: 10**

Alternate solvents: green solvents; water as a solvent; amphiphilic compounds; conventional processes and operations - current status and modification; industrial solvent selection approaches.

Unit IV: NEW DEVELOPMENTS AND INTRODUCTION TO LCA **Hours: 9**

New development processes: overview of green separation processes - distillation, chromatography, supercritical fluid extraction, membrane processes, pressurized hot water extraction, nanotechnology in separation; life cycle assessment of technology/products; industrial and research case studies in green synthesis.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand how sustainability and green chemistry are useful in chemical engineering.
CO2 - Relate the concept of green solvents to the minimization of environmental concerns.
CO3 - Utilize green chemistry principles, applications and synthesis methods to identify the application of compounds.
CO4 - Categorize conventional processes for solvent alternatives and operations.
CO5 - Recognize the importance of learning new sustainable developments in various processes.
CO6 - Develop the techniques and learn the significance of life cycle assessment applications.

TEXT / REFERENCE BOOKS

1. Garg, V. K., Yadav, A., Mohan, C., Yadav, S., & Kumari, N. (Eds.). (2023). *Green chemistry approaches to environmental sustainability: Status, challenges and prospective*. Elsevier.
2. Marteel-Parrish, A. E., & Abraham, M. A. (2013). *Green chemistry and engineering: A pathway to sustainability*. Wiley-AIChE.
3. Doble, M., & Kruthiventi, A. K. (2007). *Green chemistry and processes*. Academic Press.
4. Afonso, C. A. M., & Crespo, J. G. (2005). *Green separation processes*. Wiley-VCH.
5. Clark, J., & Macquarrie, D. (2002). *Handbook of green chemistry and technology*. Blackwell Science.
6. Atkinson, G., Dietz, S., & Neumayer, E. (2007). *Handbook of sustainable development*. Edward Elgar Publishing.